Purity in tame cohomology

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Overview

1. Motivation
2. Main Theorem: Statement and strategy
3. Tools
4. Deep dive into the proof

Notation

Throughout the presentation

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- Let X be a smooth scheme over k.
- Let Z be a smooth closed subscheme of X of pure codimension c.
- Let $i : Z \to X$ be the corresponding closed immersion.
- Let $Z\Omega^i_X = \{\eta \in \Omega^i_X | d\eta = 0\}$ be the closed differentials.

Section 1

Motivation

Gysin sequence and purity in étale cohomology

Consider a natural number $e \in \mathbb{N}$ with $(e, p) = 1$.
The Tate twist is defined by $\mathbb{Z}/e\mathbb{Z}(n) = \mu_e^{\otimes n}$. $(\mu_e := \{\alpha | \alpha^e = 1\})$
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We have a long exact sequence of étale cohomology groups

$$0 \to H^{2c}(X, \mathbb{Z}/e\mathbb{Z}(n)) \to H^{2c}(X - Z, \mathbb{Z}/e\mathbb{Z}(n)) \to H^0(Z, \mathbb{Z}/e\mathbb{Z}(n-c)) \to \dots \qquad \text{(Gysin-LES)}$$

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and isomorphisms

$$H^i(X, \mathbb{Z}/e\mathbb{Z}(n)) \cong H^i(X - Z, \mathbb{Z}/e\mathbb{Z}(n)) \quad \forall i < 2c - 1$$

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For topologists, this corresponds to <u>excision</u> for the pair $(X, X - Z)$.

Problem with p-torsion coefficients

Recall that μ_e is the sheaf of e-th roots of unity. We have a short exact sequence of sheaves

$$0 \to \mu_e \to \mathbb{G}_m \xrightarrow{(\cdot)^e} \mathbb{G}_m \to 0 \qquad \text{(Kummer SES)}$$

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If $e = p^r$, then μ_{p^r} is not an étale sheaf. However we have the following sequence

$$0 \to \mathbb{G}_m \xrightarrow{(\cdot)^{p^r}} \mathbb{G}_m \xrightarrow{\mathrm{dlog}: f \mapsto df/f} \nu_r(1) \to 0$$

where $\nu_r(1) \subseteq W_r\Omega_X$ is the logarithmic de Rham-Witt sheaf.

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where $\nu_r(1) \subseteq W_r\Omega_X$ is the logarithmic de Rham-Witt sheaf.
In a sense: $\mathbb{Z}/p^r\mathbb{Z}(n) := \nu_r(n)[-n]$. (Geisser-Levine [GL00])

log de Rham-Witt sheaves

Definition

The logarithmic de Rham-Witt sheaf $\nu_r(n)$ is the subsheaf of $W_r\Omega^n_X$ generated étale locally by elements of the form

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Example

- $\nu_r(0) \cong \mathbb{Z}/p^r\mathbb{Z}$
- $\nu_r(1) \cong \mathcal{O}_X^\times/(\mathcal{O}_X^\times)^{p^r}$

Semi-purity in étale cohomology with p-torsion coefficients

Theorem (independently by Gros [Gro85] and Milne [Mil86])

$$R^s i^! \nu_{r,X}(n) \cong \begin{cases} \nu_{r,Z}(n-c) & \text{if } s = c \\ 0 & \text{if } s \neq c, c+1 \end{cases}$$

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Corollary

We have isomorphisms $H^i(X, \nu_r(n)) \cong H^i(X - Z, \nu_r(n)) \quad \forall i < c-1$
and an injective morphism $H^{c-1}(X, \nu_r(n)) \hookrightarrow H^{c-1}(X - Z, \nu_r(n))$.

No Gysin sequence as in (Gysin-LES) since $R^{c+1} i^! \nu_{r,X}(n)$ is not necessarily zero.

$\mathbb{A}^1$ invariance in étale cohomology

Etale cohomology with coefficients in $\mathbb{Z}/e\mathbb{Z}(n)$ (e prime to p) is $\mathbb{A}^1$-invariant, i.e.

$$H^i(X, \mathbb{Z}/e\mathbb{Z}(n)) \cong H^i(X \times \mathbb{A}^1, \mathbb{Z}/e\mathbb{Z}(n)), \quad \forall i \geq 1.$$

This fails when $e = p^r \implies$ Purity fails for $X = \mathbb{P}^1$ and $Z = \{\infty\}$.

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Example

$H^1(\mathbb{A}^1_k, \mathbb{Z}/p)$ is infinite dimensional, even when k is algebraically closed.

Proof.

There are infinitely many Artin-Schreier covers $X = \mathrm{Spec}(k[x,y]/(y^p - y - f(x)))$ of $\mathbb{A}^1_k = \mathrm{Spec}(k[x])$.

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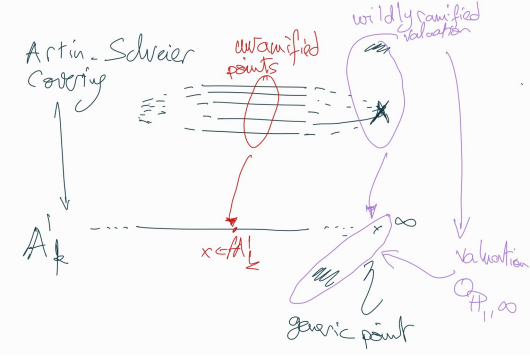

Section 2

Main Theorem: Statement and strategy

Hübner and Schmidt introduced the tame topology on schemes
$\rightarrow$ avoids wild ramification along boundaries of compactifications (e.g. Artin-Schreier covers.)

Theorem (K.)

Under the assumption of resolution of singularities (RoS), we have

$$Ri_t^! \nu_{r,X}(n) \cong \nu_{r,Z}(n-c)[-c]$$

in the tame topology.

Corollary (Merici)

Under RoS, we have $\mathbb{A}^1$-invariance in tame cohomology with coefficients in $\nu_r(n)$.

Proof idea of semi-purity in étale cohomology

(a) $R^s i^! \Omega^n = 0$ and $R^s i^! Z\Omega^n = 0$ for $s \neq c$.

(b) We have a Gysin morphism $\Omega_Z^{n-c} \xrightarrow{Gys} R^c i^! \Omega_X^n$. Similarly for $Z\Omega^n$.

(c) We have an exact sequence

$$0 \to \nu(n) \to Z\Omega^n \xrightarrow{C-1} \Omega^n \to 0 \tag{1}$$

where C is the Cartier operator ($\sim F^{-1}$).

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(d) Combine (a)-(c) to obtain the diagram

$$\begin{tikzcd}
0 \arrow[r] & \nu_Z(n-c) \arrow[r] \arrow[d, dotted, "\cong"] & Z\Omega_Z^{n-c} \arrow[r, "C-1"] \arrow[d] & \Omega_Z^{n-c} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & R^c i^! \nu_X(n) \arrow[r] & R^c i^! Z\Omega_X^n \arrow[r, "C-1"] & R^c i^! \Omega_X^n \arrow[r] & R^{c+1} i^! \nu(n).
\end{tikzcd}$$

Proof idea of purity in tame cohomology

We follow the same strategy, but

- Replace Ω^n with "logarithmic differentials" $\Omega^{+,n}$
- This requires the use of discretely ringed adic spaces instead of schemes: abstraction of schemes with boundary.

Proof idea of purity in tame cohomology

(a) $R^s i_t^! \Omega^{+,n} = 0$ and $R^s i_t^! Z\Omega^{+,n} = 0$ for $s \neq c$. (Under the assumption of RoS)

(b) We have a Gysin morphism $\Omega_{\mathcal{Z}}^{+,n-c} \xrightarrow{Gys} R^c i_t^! \Omega_{\mathcal{X}}^{+,n}$. Similarly for $Z\Omega^{+,n}$: This is an isomorphism in this case.

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\end{tikzcd}$$

Section 3

Tools

Discretely ringed adic spaces: Definition

$T \to S$ separated morphism of qcqs schemes.
$\rightarrowtail$ obtain adic space $\mathrm{Spa}(T, S)$ whose points are diagrams

$$\begin{tikzcd} & \mathrm{Spec}(k(x)) \arrow[r, "x"] \arrow[d] & T \arrow[d] \\ k(x) \text{ valuation} \arrow[r, dash] & \mathrm{Spec}(\mathcal{O}_v) \arrow[r] & S. \end{tikzcd}$$

Affinoid opens are $\mathrm{Spa}(A, A^+) \subset \mathrm{Spa}(X, S)$, fitting in the diagram

$$\begin{tikzcd} \mathrm{Spec}(A) \arrow[r, hook, "op.im."] \arrow[d] & T \arrow[d] \\ \mathrm{Spec}(A^+) \arrow[r, "f.t."] & S \end{tikzcd}$$

Let $T \to \bar{T}$ be a relative compactification of $T \to S$. Then

$$\mathrm{Spa}(T, S) \cong \mathrm{Spa}(T, \bar{T}).$$

Example

$\mathrm{Spa}(\mathbb{A}^1_k, k) \cong \mathrm{Spa}(\mathbb{A}^1_k, \mathbb{P}^1_k)$.
$\to \mathcal{O}_{\mathbb{P}^1,\infty}$ defines a point of $\mathrm{Spa}(\mathbb{A}^1_k, k)$.

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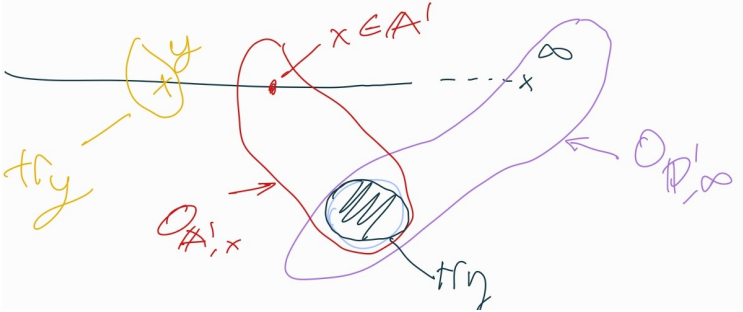
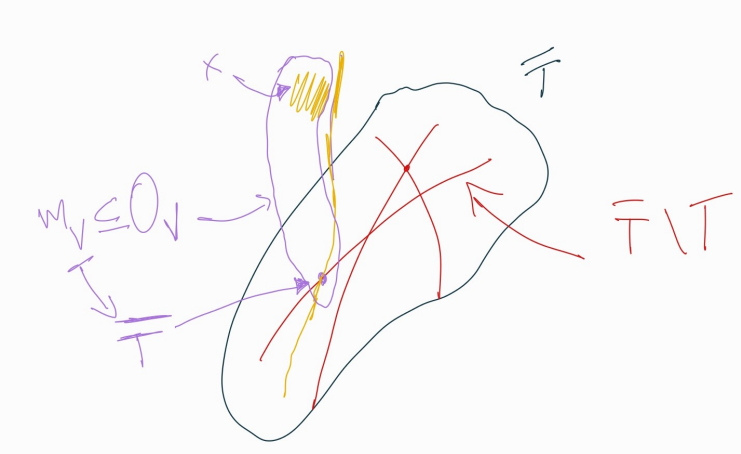


Why RoS? Part 1

Under RoS, we may assume that $\bar{T}$ is smooth over S and $\bar{T} \setminus T$ is a simple normal crossing (snc) divisor.



Tame étale morphisms and the tame site

Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a morphism of adic spaces. Then f is tame étale if

- f is étale and
- For each $x \in \mathcal{Y}$, $k(x)/k(f(x))$ is a tamely ramified extension of valued fields.

Example

The Artin-Schreier extension $k[x] \to k[x,y]/(y^p - y - x)$ is étale.
But $\mathrm{Spa}(k[x,y]/(y^p - y - x), k) \to \mathrm{Spa}(k[x], k)$ is not tame étale:
The valuation $\mathcal{O}_{\mathbb{P}^1_k,\infty}$ extends to a wildly ramified valuation on $k(x,y)$.

Notation

Denote by $\mathcal{X}_t$ the tame site of $\mathcal{X}$.

Logarithmic differentials on adic spaces: Definition

Definition

Define the sheaf $\Omega^+_{\mathcal{X}}$ as the subsheaf of $\Omega_{\mathcal{X}}$.

$$\Omega^+_{\mathcal{X}}(\mathcal{U}) := \{\eta \in \Omega_{\mathcal{X}}(\mathcal{U}) |\ \eta|_{(x,v,\epsilon)} = \sum_i f_i dg_i, |f_i g_i|_v \leq 1, \forall (x,v,\epsilon) \in \mathcal{U}\}$$

Define $\Omega^{+,n}_{\mathcal{X}} := \bigwedge^n_{\mathcal{O}^+_{\mathcal{X}}} \Omega^+_{\mathcal{X}}$. (e.g. $\Omega^{+,0}_{\mathcal{X}} = \mathcal{O}^+_{\mathcal{X}}$)

Proposition (Hübner)

$\Omega^{+,n}_{\mathcal{X}}$ is a tame étale sheaf. But not an étale sheaf in general.

Logarithmic differentials on adic spaces: General facts

- For each valuation $|df/f|_v \leq 1 \implies \nu(n) \subseteq \Omega^{+,n}_{\mathcal{X}}$.
- Let (A, A^+) be a "local Huber pair". Then

$$\Omega^{+,n}_{(A,A^+)} = \Omega^{+,n}_{A^+}(\log(A^\times \cap A^+)) =: \Omega^{n,\log}_{(A,A^+)}$$

Why RoS? Part 2

If $\mathcal{U} = \mathrm{Spa}(U, \bar{U})$ with $(U, \bar{U})$ log smooth, then

$$\Omega^{+,n}_{\mathcal{X}}(\mathcal{U}) = \Omega^n_{\overline{U}}(\log(\overline{U} - U)).$$

This is not true for non (log-)smooth pairs $(U, \bar{U})$.

Section 4

Deep dive into the proof

Setting

- $\mathcal{X} := \mathrm{Spa}(X, k)$.
- $i : \mathcal{Z} := \mathrm{Spa}(Z, k) \to \mathcal{X}$.
- $j : \mathcal{U} := \mathrm{Spa}(X - Z, k) = \mathcal{X} - \mathcal{Z} \to \mathcal{X}$.
- $i_t^! : \mathrm{Sh}(\mathcal{X}_t) \to \mathrm{Sh}(\mathcal{Z}_t)$ is the functor defined by

$$\mathcal{F} \mapsto \ker(\mathcal{F} \to j_* j^* \mathcal{F}).$$

Goal

Prove the purity isomorphism

$$R i_t^! \nu_{r,\mathcal{X}}(n) \cong \nu_{r,\mathcal{Z}}(n - c)[-c].$$

Recall: Proof idea of purity in tame cohomology

(a) $R^s i_t^! \Omega^{+,n} = 0$ and $R^s i_t^! Z\Omega^{+,n} = 0$ for $s \neq c$. (Under the assumption of RoS)

(b) We have a Gysin morphism $\Omega_{\mathcal{Z}}^{+,n-c} \xrightarrow{Gys} R^c i_t^! \Omega_{\mathcal{X}}^{+,n}$. Similarly for $Z\Omega^{+,n}$: This is an isomorphism in this case.

(c) We have an exact sequence

$$0 \to \nu(n) \to Z\Omega^{+,n} \xrightarrow{C-1} \Omega^{+,n} \to 0$$

(d) Combine (a)-(c) to obtain the diagram

$$\begin{tikzcd}
0 \arrow[r] & \nu_{\mathcal{Z}}(n-c) \arrow[r] \arrow[d, dotted, "\cong"] & Z\Omega_{\mathcal{Z}}^{+,n-c} \arrow[r, "C-1"] \arrow[d, "\cong"] & \Omega_{\mathcal{Z}}^{+,n} \arrow[r] \arrow[d, "\cong"] & 0 \\
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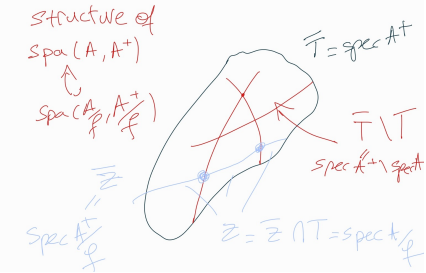
\end{tikzcd}$$

Deep dive into step (b): The Gysin isomorphism

Proof idea (Hübner,K.)

- Reduce to codimension $c = 1 \implies Z = V(f)$ for some $f \in \mathcal{O}_X$.
- Reduce to local Huber pairs (A, A^+)
  $\implies \mathcal{U} = \mathrm{Spa}(A[f^{-1}], A^+), \mathcal{X} = \mathrm{Spa}(A, A^+)$ and $\mathcal{Z} = \mathrm{Spa}(A/f, A^+/f)$.



Proof idea (cntd.)

- Excision $\implies$ Explicit description $R^1 i^! \Omega^{+,n}_{\mathcal{X}} \cong j_* \Omega^{+,n}_{\mathcal{U}} / \Omega^{+,n}_{\mathcal{X}}$
- We have a residue short exact sequence

$$0 \to \Omega^{n,\log}_{(A,A^+)} \to \Omega^{n,\log}_{A^+}(\log A[f^{-1}]^\times \cap A^+) \xrightarrow{res} \Omega^{n-1,\log}_{(A/f,A^+/f)} \to 0.$$

$$\mathrm{dlog} f \wedge \omega \mapsto \overline{\omega}$$

A short explanation of step (c)

Assumg RoS, the restriction of $C : Z\Omega^n_{\mathcal{X}} \to \Omega^n_{\mathcal{X}}$ to $Z\Omega^{+,n}_{\mathcal{X}}$ factors through $\Omega^{+,n}_{\mathcal{X}}$ (argument by logarithmic differentials on smooth pairs).

Proof of (c) assuming that C is well-defined on Ω^+.

We only have to check exactness on the right. Assume that $\mathcal{X} = \mathrm{Spa}(A, A^+)$. Consider $adb \in \Omega^+(\mathrm{Spa}(A, A^+))$ such that $|ab|_v \leq 1$ for each $v \in \mathrm{Spa}(A, A^+)$. The preimage of $adb = (ab)d\log b$, where $ab \in A^+$ is found by solving $X^p - X - ab$. □

It is subtle to show that $|C(\eta)|_v \leq 1 \implies |\eta|_v \leq 1$. (use logarithmic differentials on valuation rings)

Putting everything together, we obtain the purity isomorphism

Theorem (K.)

Under the assumption of resolution of singularities (RoS), we have $Ri_t^! \nu_{r,X}(n) \cong \nu_{r,Z}(n-c)[-c]$ in the tame topology.

Setting

$$H_t^i(X, \nu_r(n)) = H^i(\mathrm{Spa}(X,k)_t, \nu_r(n)),$$

we obtain a long exact sequence

$$0 \to H_t^0(Z, \nu_r(n-c)) \to H_t^c(X, \nu_r(n)) \to H_t^c(X-Z, \nu_r(n)) \to H_t^1(Z, \nu_r(n-c)) \to \dots$$

Thank you for your attention!

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